Akoizo Original Practice - 2027 Rotation Form 3

Division C | Chemistry Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) At constant temperature, a gas at 1.6 atm occupies 7 L. Compress it to 3.5 L. Find the final pressure.

Answer: _____
_____
_____

2. (2 points) At constant pressure, a gas occupies 7 L at 360 K. Find its volume at 720 K.

Answer: _____
_____
_____

3. (2 points) A rigid vessel contains a gas at 1.6 atm and 360 K. Find its pressure at 180 K with no gas lost.

Answer: _____
_____
_____

4. (2 points) Use PV=nRT with R=0.08206 L atm/(mol K). P=1.6 atm, V=7 L and T=360 K. Find the amount of gas.

Answer: _____
_____
_____

5. (2 points) A dry gas mixture has partial pressures 1.6 atm and 0.8 atm. Find total pressure.

Answer: _____
_____
_____

6. (2 points) A mixture has 5 mol oxygen and 15 mol nitrogen at total pressure 7 atm. Find oxygen partial pressure.

Answer: _____
_____
_____

7. (2 points) Gas collected over water has total pressure 103 kPa. Water-vapor pressure is supplied as 6 kPa. Find dry-gas pressure.

Answer: _____
_____
_____

8. (2 points) Two ideal gases at the same temperature have molar masses 4 and 16 g/mol. What is the diffusion-rate ratio light/heavy? Use Graham's law.

Answer: _____
_____
_____

9. (2 points) For 2 H2 + O2 -> 2 H2O, 5 mol O2 reacts completely with excess H2. How many moles of H2 are consumed?

Answer: _____
_____
_____

10. (2 points) Explain why gas-law temperature ratios must use kelvin rather than degrees Celsius.

Answer: _____
_____
_____

11. (2 points) [State/National] Determine the order with respect to A from the supplied initial-rate data.

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

Answer: _____
_____
_____

12. (2 points) [State/National] Determine the order with respect to B from the supplied initial-rate data.

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

Answer: _____
_____
_____

13. (2 points) [State/National] Calculate k from the initial-rate data. Give its value in M^-2 s^-1.

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

Answer: _____
_____
_____

14. (2 points) [State/National] Predict the rate at [A]=0.48 M and [B]=0.10 M.

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

Answer: _____
_____
_____

15. (2 points) [State/National] For a first-order reaction, k=0.05330769 s^-1. Use ln(2)=0.693 to calculate the half-life.

Answer: _____
_____
_____

16. (2 points) [State/National] A first-order reactant starts at 1.04 M. Find concentration after three half-lives.

Answer: _____
_____
_____

17. (2 points) [State/National] A zero-order reaction starts at 1.3 M and consumes reactant at 0.02 M/s. Find concentration after 10 s.

Answer: _____
_____
_____

18. (2 points) A catalyst increases rate without changing reactant/product free energies. Explain the mechanism and whether the equilibrium constant changes.

Answer: _____
_____
_____

19. (2 points) Equal masses of the same solid react with excess acid under matched conditions. Why can smaller particles react faster?

Answer: _____
_____
_____

20. (2 points) [State/National] At 300 K a rate constant is k1; at 330 K it is 2k1. Using R=8.314 J/(mol K), ln(2)=0.693 and ln(k2/k1)=Ea/R(1/T1-1/T2), find Ea.

Answer: _____
_____
_____